Function-Driven Topology Optimization of a Dual-Purpose Robotic Tooling Arm

Engineering pilot study, documentation revised 25 October 2026

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Status: pilot comparison complete, simulation only. No part was printed or tested.

Abstract

A single-piece aluminium tooling arm, carrying one robot interface and two tool interfaces, was topology optimized three times in nTop 6.0.3: for a handling load only, for a rotary load (torque with axial thrust) only, and for both load cases kept separate inside one weighted objective. Each density result was rebuilt as a solid, remeshed and analysed under both loads, so that every design was also tested under the load it was not optimized for. Together with the unoptimized baseline this gives eight static cases. All three optimized arms end between 1143 g and 1163 g, about 47 % lighter than the 2183 g baseline.

The clearest result is an asymmetry between the two specialists. Optimized for bending only, the handling arm has 29.66 times the dual-mode arm's estimated compliance under rotary loading, and its fitted tool-flange rotation is 0.6107° against 0.02126°. Under the handling load, the rotary arm is 18.9 % more compliant than the handling specialist. The dual-mode arm lies between the specialists: its estimated compliance is 10.49 % higher than handling-only under A and 8.21 % higher than rotary-only under B.

The mean normalized external-work estimates differ by 0.65 % between rotary-only and dual-mode, while their reconstructed masses differ by 0.54 %. Final-body mesh sensitivity and exact work-conjugate compliance were not established, so their overall ranking remains unresolved. This is a comparative simulation study, not a strength qualification or AM feasibility assessment.

1. Engineering question and hypotheses

The study asks how optimizing one mechanical component for two independent functions changes its material distribution and structural response compared with optimizing the same component for only one function. The two tasks never act at the same time. The arm was chosen because one robot interface and two distinct tool interfaces give the load cases separate paths through a shared body. Multi-function end-of-arm tooling exists in industry; one published case uses a binder-jet printed adapter to carry two tools at 90° on a robot arm for automotive inspection [1]. That example motivates the application only. The dimensions, mounting pattern and loads here are independent benchmark assumptions, not values from that product.

Before evaluating the results, the engineering expectations were: handling-only optimization would favour bending stiffness; rotary-only would favour torsional stiffness; the weighted dual-mode objective would seek a compromise; and a design optimized without one load might perform poorly when cross-tested under that load. The experiment tests those expectations rather than presuming the dual-mode design must win.

Multi-load compliance minimization with weighted objectives is an established method [2]; this project does not claim a new algorithm. The intended control is one baseline, one material model, common protected regions, design envelope, FE model and optimization budget, with only the objective changed. The common settings are listed in Table 5 and in config/engineering_inputs.json. Each reconstructed design is then evaluated under both load cases, including the one its optimization ignored.

2. Component, material and load cases

Table 1. Baseline geometry (mm). Origin at the centre of the rear mounting face, +X from robot to tool, +Z vertical.

Feature	Geometry	Role
Robot plate	80 × 80 × 10, x = 0 to 10	Robot interface, rear face fixed
Root holes	4 × Ø6.6 at (y, z) = (±28, ±28)	Mounting clearance
Root boss	Ø50, x = 10 to 30	Transition into the arm
Design envelope	x = 25 to 225, y = ±32, z = ±25	Material that may be removed
Handling pad	Ø52 at (x, y) = (185, 0), z = 20 to 35	Handling tool face at z = 35
Pad holes	4 × Ø4.2 blind, z = 35 to 25	M5 tap drill representation
Rotary hub	Ø40, x = 215 to 230	Torque path into the flange
Rotary flange	Ø80, x = 230 to 240	Rotary tool face at x = 240
Flange holes	4 × Ø5.5 on PCD 60	Tool fastener clearance

The parameter-driven Fusion v02 model was exported as cad/baseline/tooling_arm_baseline_v02.step. A prior Fusion reimport check found one solid with 36 faces, 59 edges, a 240 × 80 × 80 mm bounding box and 808,365.279 mm³ B-rep volume. The STEP file in this project matches the supplied download by SHA-256. These CAD checks establish geometry and scale, not the nTop FE setup. The Fusion archive assigned a steel physical material (7850 kg/m³); the simulations and masses below use the separately assigned aluminium benchmark. No steel result is used in the comparison.

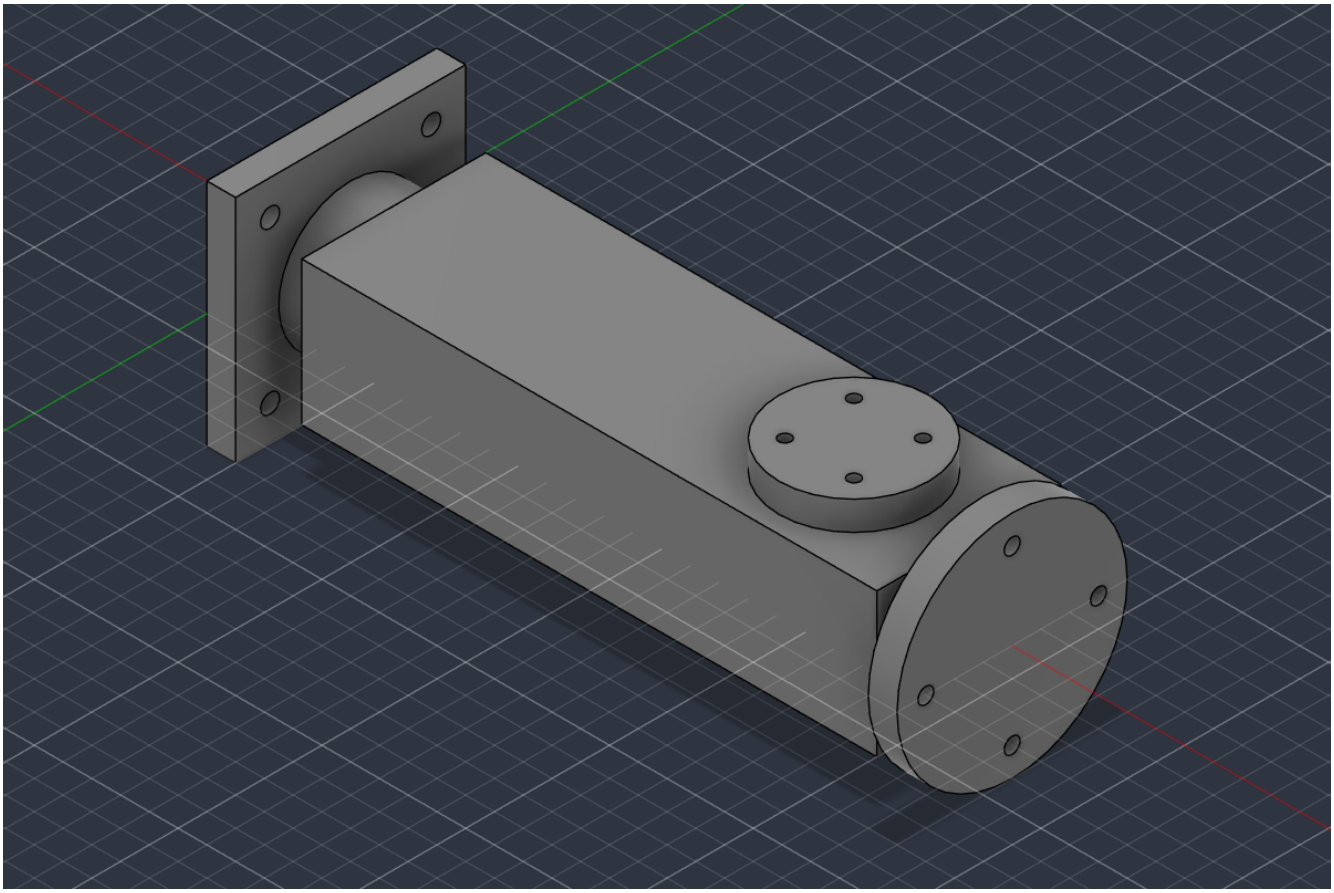


Figure 1. Original Fusion STEP reimport, before topology optimization. Source: saved Fusion review image. The view shows the three mechanical interfaces but does not by itself verify an FE load selection.

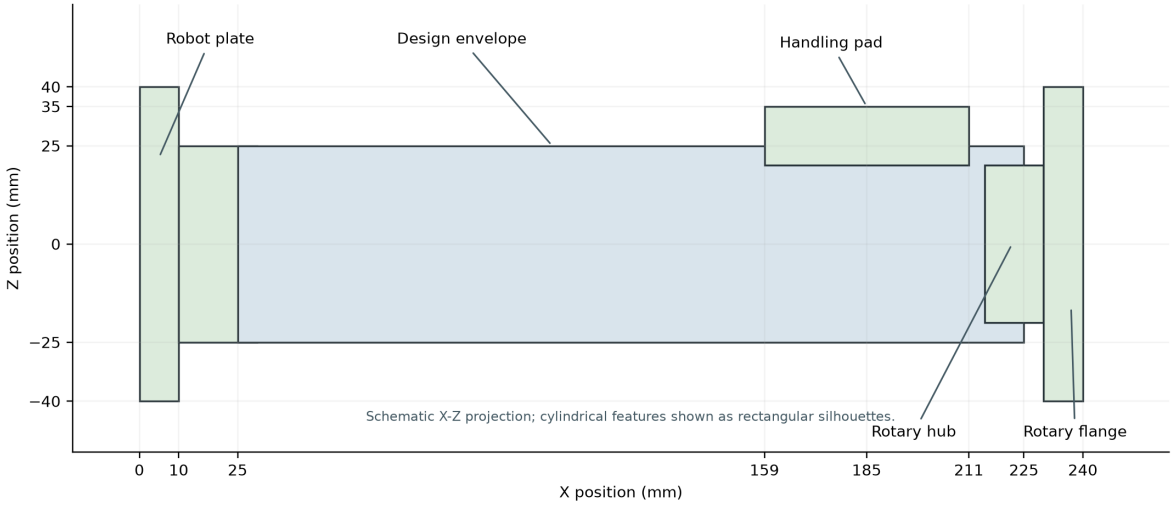


Figure 2. Technical X–Z schematic derived from the specified dimensions in Table 1. Cylindrical features are shown as rectangular silhouettes; the image is not a section of the CAD B-rep.

Table 2. Load cases. Both use the same restraint: all translations fixed on the rear face at x = 0.

Case	Applied load	Location	Expected root reaction
A, handling	F = (0, 0, −250) N	Pad top face, z = 35	Fz = +250 N, My = −46.25 N·m
B, rotary	F = (−300, 0, 0) N and M = (+24, 0, 0) N·m	Flange end face, x = 240	Fx = +300 N, Mx = −24 N·m

The material is an isotropic linear elastic aluminium benchmark (E = 70 GPa, ν = 0.33, ρ = 2700 kg/m³). It is not a qualified AM alloy, and no yield or fatigue allowable was assigned. Placing the two tool faces at separate locations gives each load its own path to the root, which is what allows a single-mode design to sacrifice the path it does not need.

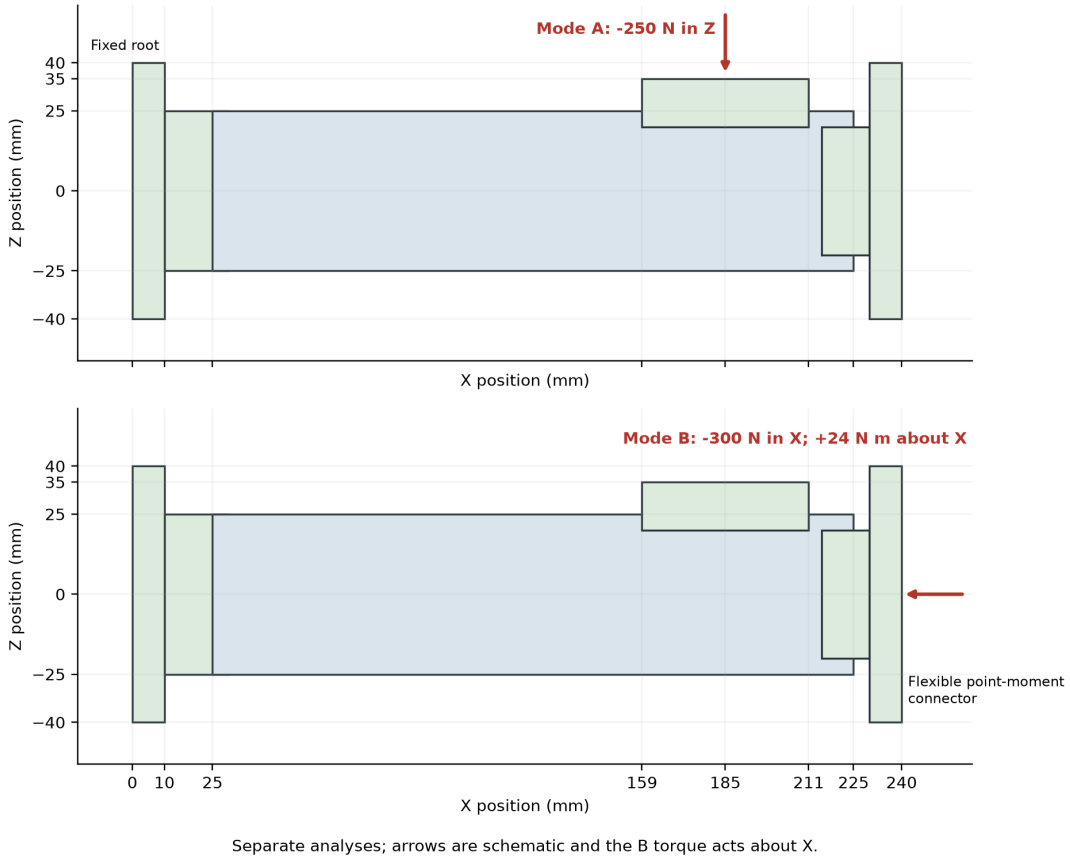


Figure 3. Separate, mutually exclusive load cases on a schematic X–Z projection. A applies −250 N at the pad top; B applies −300 N axially and +24 N·m about X at the distal flange. Arrows are schematic, and the rear face is fixed in translation. Source: Table 2 and the Stage 1 specification.

The recorded implementation applies torque through nTop's Point Moment block with a flexible connector over the flange end. Flexible RBE3-type connectors distribute loading while allowing the selected face to deform, avoiding the local rigid coupling of an RBE2-type connector [3]. The original specification proposed four 200 N tangential bolt-region forces on the 60 mm pitch circle; that bearing model was not run. Across the four case B analyses, the torque term supplies 93.7–96.7 % of the estimated work, so B is predominantly a torsion test with a smaller axial contribution.

Normalizing each compliance by the baseline response under the same load removes a uniform scale change to that complete load case in a linear model. The relative thrust-to-torque ratio within B still changes its load path and response. The specific load magnitudes are benchmark choices, not qualified tool requirements.

3. Design space and material budget

Protected solids were built from primitives and intersected with the baseline body, so that they keep the interface material without filling any holes: P1 covers the plate and root boss, P2 the handling pad and a 10 mm land below it, and P3 the hub and flange.

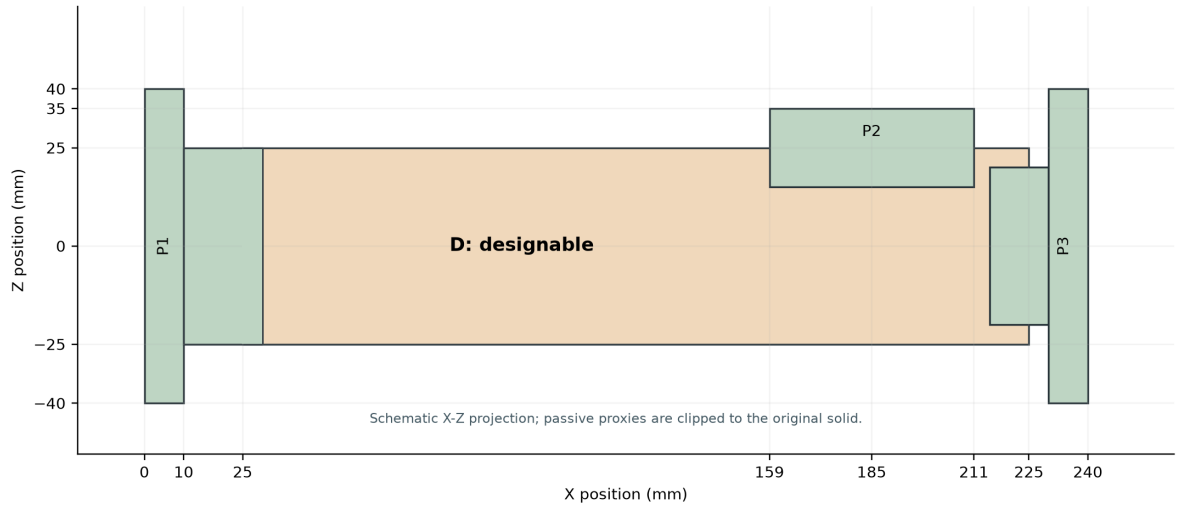


Figure 4. Schematic X–Z projection of protected P1–P3 and designable D. P1 covers the plate and the root boss up to x = 30 mm, so it overlaps the start of the envelope. The protected proxy solids were clipped to the baseline before union so existing holes stayed open. Schematic only, not an nTop region plot.

Table 3. Volume accounting from nTop Mass Properties.

Quantity	Volume (mm³)	Mass at 2700 kg/m³ (g)
Baseline body V0	808 438.0	2182.78
Preserved regions VP	211 905.9	572.15
Designable region VD = V0 − VP	596 532.1	1610.64
Target: VP + 0.35 VD	420 692.1	1135.87

The budget keeps 35 % of the designable volume while all preserved material stays in place. Algebraically, Vtarget = VP + 0.35(V0 − VP) = 420,692.1441 mm³ and f = Vtarget/V0 = 0.5203765. At 2700 kg/m³ this is 1135.8688 g. The Volume Fraction Response was scoped to an FE region covering the full baseline body, so preserved cells count in both numerator and denominator, and it was limited to 0.52038. A separately measured designable value of 596,404.3578 mm³ is 127.7429 mm³ below V0 − VP. Its cause is untraced, so the calculation retains the algebraic value and discloses the difference. The nominal density-domain constraint and final thresholded-solid masses must be judged separately.

4. Baseline analysis and checks

The full-solid baseline was solved with linear tetrahedra at two mesh sizes before any optimization. This established the response scales, checked the intended bending and torsional resultants, and screened global mesh sensitivity before changing the material layout.

Table 4. Baseline mesh comparison.

Quantity	4.0 mm mesh	2.5 mm mesh	Change
Nodes	23 244	71 589	
Compliance, case A (J)	0.003907	0.003969	+1.59 %
Compliance, case B (J)	0.005034	0.005154	+2.38 %
Maximum displacement A (mm)	0.02234	0.02277	+1.94 %
Maximum displacement B (mm)	0.008295	0.008493	+2.39 %
Peak von Mises A (MPa)	3.251	3.860	+18.7 %
Peak von Mises B (MPa)	4.492	4.617	+2.8 %

The displacement-derived global work estimates change by less than 2.5 % between these two meshes, which was accepted for this pilot; the 2.5 mm baseline mesh (396,426 elements, 71,589 nodes) was then used for all three optimizations. This is a two-point screening comparison, not proof of convergence. The A peak changes by 18.7 %, so local peak stress cannot be treated as converged. The reported peak locations are near the plate-to-boss and hub-to-flange transitions, where sharp geometry also limits a nodal-maximum interpretation.

Root reactions were exported for every node and summed independently. On the 2.5 mm baseline they reproduce the principal applied resultants to within about 0.0004 N and 0.0005 N·m. The complete eight-case principal reaction sums appear in Appendix B. Equilibrium checks the applied resultants and numerical solution balance; it does not establish mesh convergence or realistic joint loading.

5. Topology optimization setup

The setup follows nTop's documented topology optimization workflow [4]: FE model, boundary conditions, compliance response as objective, volume fraction response as constraint.

Table 5. Settings shared by all three optimizations.

Setting	Value
FE model	2.5 mm linear tetrahedra, 71 589 nodes
Constraints	Preserved regions solid; volume fraction ≤ 0.52038
Filter size	8 mm
Maximum iterations	100
Minimum objective change	0.0005
Minimum density change	0.01
Boundary penalty	0.5
Iterations run (handling, rotary, dual)	37, 38, 38

The handling arm minimizes C_A, and the rotary arm minimizes C_B. The dual-mode arm minimizes w_A·C_A + w_B·C_B, where C_A and C_B are two separate Structural Compliance Responses [5], each with its own boundary-condition list. Combining both loads into one simultaneous case would produce a different energy expression, including cross terms, and answer a different question [2].

The objective histories end after 37, 38 and 38 iterations for handling, rotary and dual (Figure 11). At the stop, the maximum design-variable change was about 0.13 for handling and 0.08 for rotary, above the 0.01 setting. The solver's stopping reason and final constraint residual were not recorded, so a flat objective history is not treated as proof of convergence. The rotary-only run was repeated after a first attempt used a different filter size and save increment; all rotary results in this report come from the repeated run with the common settings.

5.1 Choosing the dual-mode weights

The two compliances are not on the same scale, and their size depends on the load magnitude, since compliance grows with the square of the load. A plain sum C_A + C_B would let whichever case happens to be larger steer the design, and doubling one load would quietly change the balance between the two functions. nTop does not normalize the weights of a multi-response objective and recommends weighting the responses to a similar order of magnitude [6]. The study therefore uses the normalized form

$$J = 0.5 \cdot C_A / C_{A0} + 0.5 \cdot C_B / C_{B0},$$

where C_A0 and C_B0 are the compliances of the solid baseline under each case. Written as nTop weights, this is w_A = 0.5 / C_A0 and w_B = 0.5 / C_B0. The equal factor of 0.5 is a decision, not a result: it states that both functions matter equally, and a different split would select a different point on the trade-off between them.

Table 6. How the dual-mode weights were derived and checked.

Basis	C_A ref (J)	C_B ref (J)	w_A	w_B	w_B / w_A
Iteration 1 of the single-mode runs (used)	0.05145	0.06782	9.718	7.372	0.7586
Solid baseline, external work (check)	0.003969	0.005154	125.98	97.01	0.7701
Solid baseline, case B without torque work	0.003969	0.000183	125.98	2738.5	21.74

When the dual-mode run was set up, nTop's static-analysis result offered displacement, strain, stress and reaction fields but no compliance or strain-energy value for the solid model [7]. The pilot weights were instead taken from the first displayed iteration of the handling-only and rotary-only runs. Those first-iteration values are not the full-solid C_A0 and C_B0. Their ratio was an expedient estimate of relative response scale, to be checked after baseline displacement exports became available.

Once the eight reanalyses were complete, the solid baseline external-work estimates were calculated from exported displacements (Section 8). The iteration-1 values are about 12.96 and 13.16 times those proxies, and the pilot ratio 0.7586 differs by about 1.5 % from the baseline-derived ratio 0.7701. A common multiplicative scaling of both weights leaves the relative emphasis unchanged. The ratio comparison does not prove equality between nTop's structural-compliance response and the Python proxy, nor does it establish identical optimizer termination under a rescaled objective.

The third row of Table 6 shows why case B compliance must include the torque work. With only the axial force term, C_B0 drops to 0.000183 J and the weight ratio rises to 21.7, which would have made the rotary function about 29 times more important than intended. An intermediate estimate made this mistake before it was caught. In the solid baseline the torque supplies 96.5 % of the case B work, so leaving it out does not give a rough answer but a wrong one.

The first-iteration pilot references put the two weighted contributions near 0.5 each, so the displayed dual objective starts near 1.0 and has no physical unit. The pilot weights express the intended balance for the chosen load cases; a change in the thrust-to-torque ratio would change that balance. The final comparison in Section 7 uses the same form of two-case normalization but uses solid-baseline displacement-derived proxies rather than the optimizer's exact responses and first-iteration references. It is an evaluation of the intended trade-off, not an identical replay of the optimizer objective.

6. Reconstruction and final meshes

The recorded reconstruction chain was density output → Implicit Body from Topology Optimization at threshold 0.5 → Smoothen Body → union with the preserved regions → intersection with the baseline body. The last step clips material outside the original envelope; the documented handling mass fell from 1149.6107 g before intersection to 1143.2641 g afterward. The chain follows nTop's documented post-processing [4]. Its example blends the union with a 2 mm radius; this study used a zero radius for all three designs, which keeps the interfaces exact but leaves sharp transitions where optimized and protected material meet. The dual-mode smoothing used a 1 mm grid and 3 smoothing iterations; the same settings were intended for the other two designs, but their smoothing inputs were not recorded side by side.

Table 7. Reconstructed designs.

Design	Volume (mm^3)	Mass (g)	Above target	Mass saved vs baseline
Handling-only	423 431.1	1143.26	+0.65 %	47.62 %
Rotary-only	430 669.9	1162.81	+2.37 %	46.73 %
Dual-mode	428 357.6	1156.57	+1.82 %	47.01 %

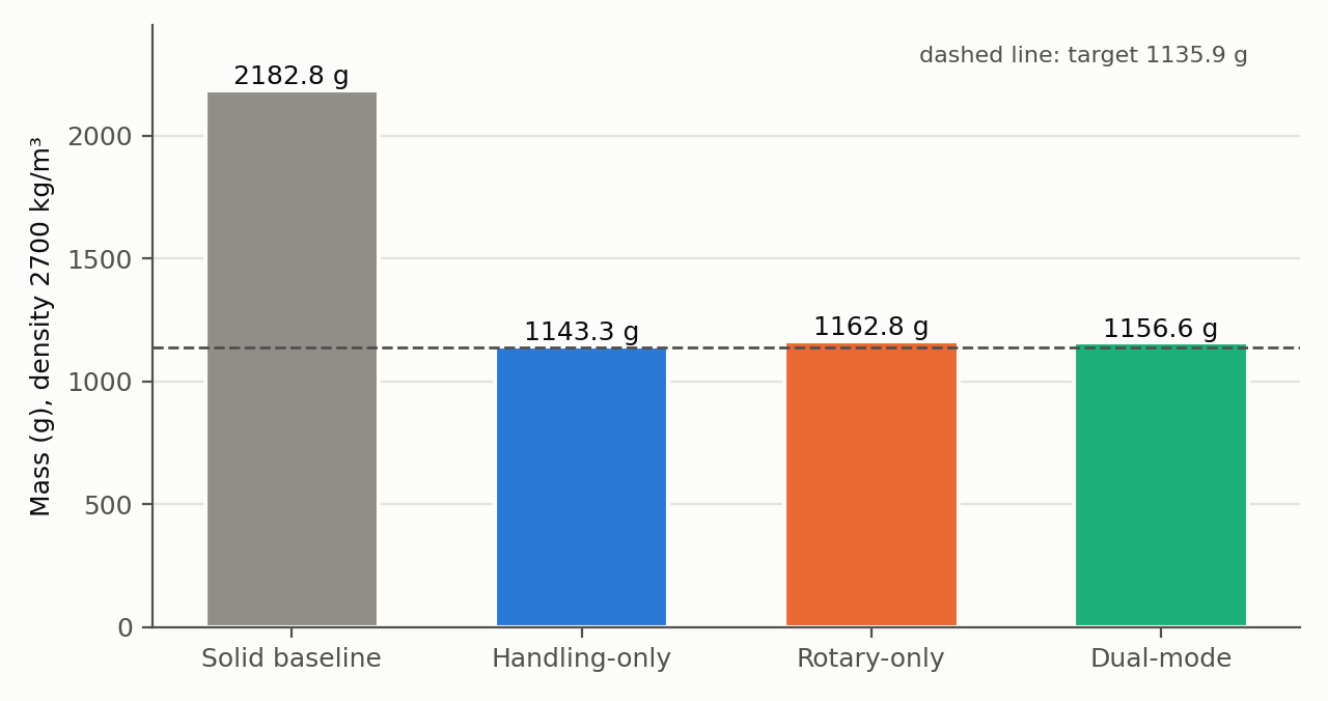


Figure 5. Python-generated mass comparison using nTop Mass Properties values recorded in `config/engineering_inputs.json`. The dashed line is the 1135.9 g target; none of the reconstructed solids reaches it.

All three reconstructed bodies exceed the target, and by different amounts. The final rotary arm is 6.24 g heavier than the final dual arm. The optimization density-volume target therefore did not become an equal-mass comparison after thresholding, smoothing and clipping. The source of each mass difference within the reconstruction chain was not quantified separately.

Direct FE Volume Mesh attempts on the reconstructed implicit bodies produced self-intersecting elements; an attempted 1.5 mm setting did not remove the warning, so those meshes were rejected. The accepted route uses Mesh from Implicit Body at 0.75 mm surface tolerance, Remesh Surface to 2.5 mm triangles, Robust Tetrahedral Mesh at 2.5 mm with regularity 10, then conversion to a linear FE mesh. The 0.75 mm value controls surface approximation, not FE element size. The final meshes have 52,635 (handling), 55,371 (rotary) and 55,073 (dual) nodes; the recorded dual mesh has 267,810 tetrahedra. The accepted meshes have no intersecting elements, a split check finds one connected surface component per body, and the maximum skewness is about 0.970, 0.978 and 0.975 respectively. Low minimum quality is recorded, including 0.085 for dual. These checks do not establish final-body displacement or stress convergence, local thickness adequacy, or mesh-to-CAD fidelity.

New load and restraint regions were selected on each final mesh with 0.2 mm-thick boxes at root x=0, pad z=35 and flange x=240. In the exported node coordinates, the remeshed interface planes sit 0.06 to 0.08 mm inside the nominal planes on average, so the boxes capture about 85 % of the nodes lying within 0.75 mm of the root plane, 77 to 78 % for the pad and 84 % for the flange. These are node fractions, not measured face areas. Each final model needs boundaries tied to its own FE mesh. Mesh-reference errors in the handling and rotary models were corrected before the accepted solves; the current full-vertex exports do not show the earlier one-row reaction-file symptom.

7. Results

Table 8a. Kinematics and displacement-derived external-work estimates for eight static cases, grouped by geometry. Pad uz is the mean selected-node vertical displacement under A; flange ux and fitted twist use selected nodes under B. The work values are not native solver-integrated compliance.

Design	Pad uz A (mm)	Flange ux B (mm)	Twist B (°)	C_A estimate (J)	C_B estimate (J)
Solid baseline	−0.01588	−0.000609	0.01187	0.003969	0.005154
Handling-only	−0.02165	−0.05745	0.6107	0.005412	0.27303
Rotary-only	−0.02573	−0.000978	0.01961	0.006433	0.008507
Dual-mode	−0.02392	−0.001004	0.02126	0.005980	0.009205

Table 8b. Stress from the same eight static cases. Peak is the nodal maximum von Mises value; p99.9 is an unweighted nodal percentile, not a converged material-volume statistic.

Design	Peak A / B (MPa)	p99.9 A / B (MPa)
Solid baseline	3.86 / 4.62	3.05 / 3.52
Handling-only	4.26 / 485.4	3.35 / 162.3
Rotary-only	4.09 / 4.98	3.41 / 4.11
Dual-mode	4.61 / 4.92	3.39 / 3.98

Root reactions balance the principal applied resultants in all eight cases: the force sums are within 0.001 N of 250 N and 300 N, the case B root Mx is −24.000 N·m to displayed precision, and case A root My lies between −46.215 and −46.250 N·m. Small transverse reaction moments also occur and are listed in the source table. The A moment difference is consistent with the selected remeshed pad centroid shifting from nominal x=185 mm; that causal explanation was not independently checked against a load-distribution export.

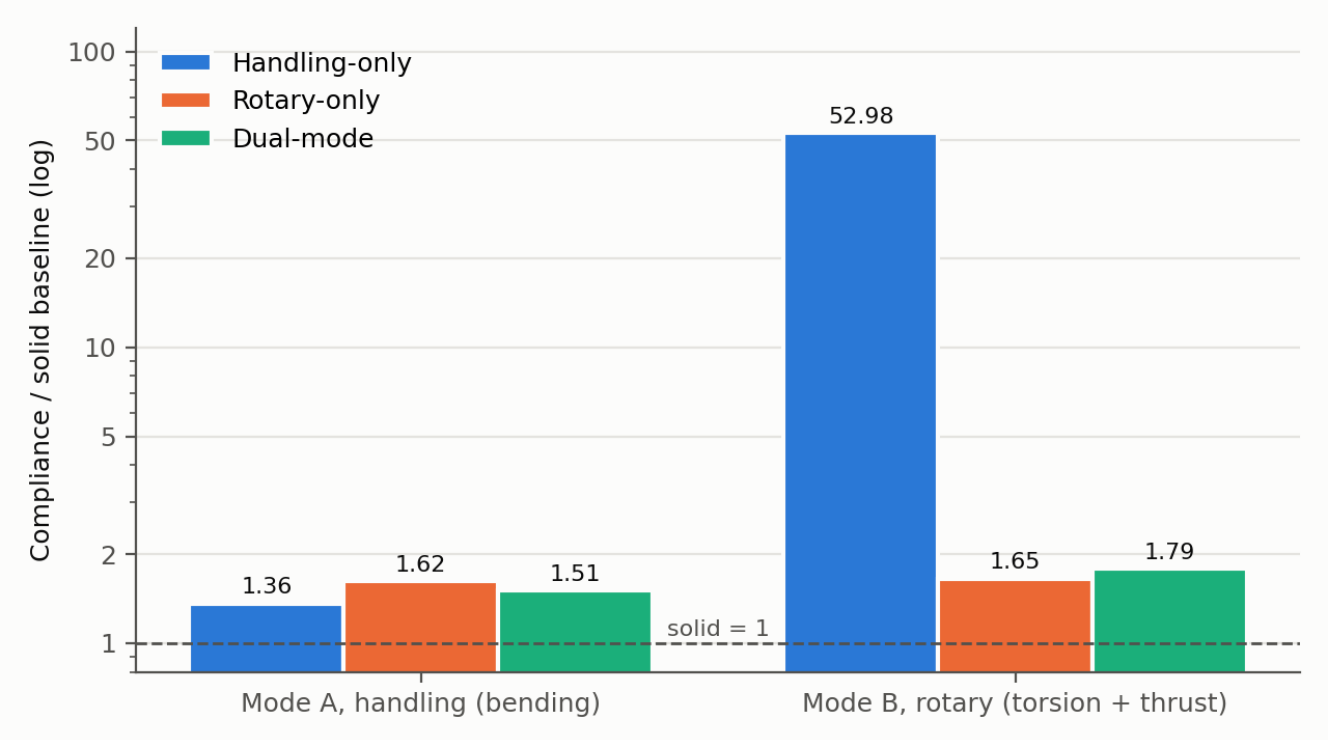


Figure 6. Python-generated comparison of external-work estimates divided by the corresponding solid-baseline estimate. Logarithmic vertical scale; the bars are calculated from nTop displacement exports, not native compliance-response values.

Under the rotary load, the handling arm's estimated compliance is 52.98 times the baseline value and 29.66 times that of the dual-mode arm. Its fitted flange rotation is 0.6107°, compared with 0.02126° for dual. A narrow connection ahead of the hub is visible in the saved nTop view, and the B nodal-stress maximum occurs near x=213.4 mm. Rotary-only remains much closer to the other lightweight designs under A. One plausible

explanation is that material arranged around the arm axis to resist torsion also carries bending, whereas the handling layout leaves a weak torsional connection. Cross-sections and section properties were not measured; this is an interpretation of the observed layouts and displacements, not a demonstrated general rule.

The handling B linear-elastic nodal peak is 485.4 MPa at the narrow connection. Only 16 nodes exceed 300 MPa and the 99.9th nodal percentile is 162.3 MPa. These numbers flag a local strength concern but do not predict yield or failure: the peak is not mesh-converged, no production alloy or AM process was specified, and the joint/load idealizations remain. A published as-built LPBF AlSi10Mg coupon value around 250 MPa [8] is context only and is not an allowable for this benchmark. All other present nodal peak values are below 5 MPa at the illustrative loads. The current pilot compares global structural response, not qualified strength.

Table 9. Normalized comparison using each solid-baseline external-work estimate as C0; lower values mean less estimated deformation work under the specified load.

Design	Mass (g)	C_A / C_A0	C_B / C_B0	Mean score	Worst case
Handling-only	1143.26	1.364	52.98	27.17	52.98
Rotary-only	1162.81	1.621	1.651	1.636	1.651
Dual-mode	1156.57	1.507	1.786	1.646	1.786

Against the specialists, the dual-mode arm has 10.49 % higher estimated compliance than handling-only in A and 8.21 % higher than rotary-only in B. These are observed differences at the realized unequal masses; they should not be presented as equal-mass penalties caused only by adding a second objective.

Rotary-only and dual-mode cannot be separated on the mean score at the current level of validation. The rotary value is numerically 0.65 % lower, but the rotary body is also 6.24 g (0.54 %) heavier. A purely illustrative scaling C □ m^−k, applied to the measured masses and proxy scores, crosses at k≈1.20 (Figure 7). No exponent was fitted and no mass-matched FE models were solved; the plot shows sensitivity, not a corrected result.

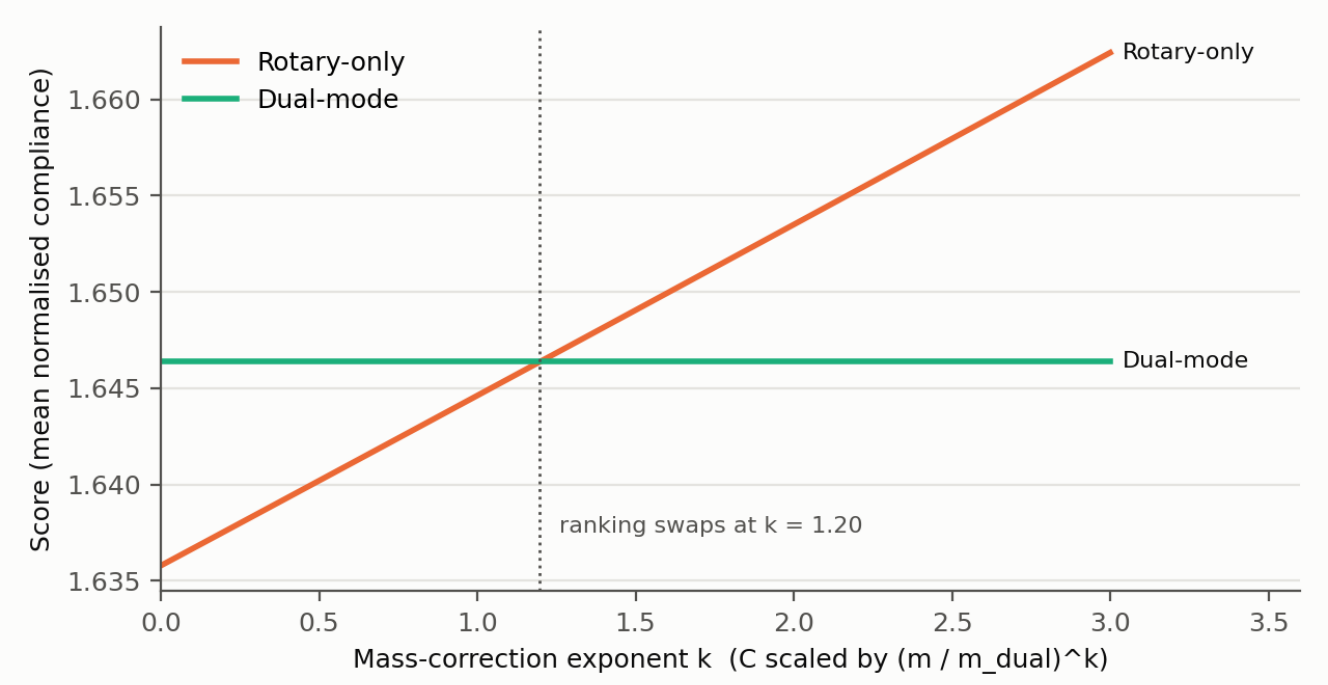


Figure 7. Python-generated hypothetical mass sensitivity. C is scaled by (m/m_dual)^k; k is an assumed exponent, not a fitted material or topology law. The curves are not mass-matched simulation results.

The rotary design also has the lower observed worst-case normalized proxy (1.651 versus 1.786), although worst-case compliance was not the objective optimized here. Applying the pilot weights to the final external-work proxies gives 0.12524 for rotary and 0.12598 for dual, a 0.59 % numerical gap. The optimizer operated on density fields, whereas this comparison uses thresholded bodies, unequal masses, different

meshes and proxy compliance; neither result proves that the dual optimization failed. The present evidence cannot apportion the small gap among reconstruction, mesh, response definition and optimization history.

The topology screenshots below come from the nTop notebook.

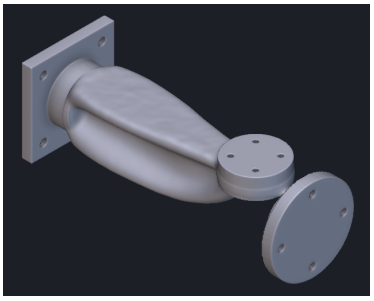
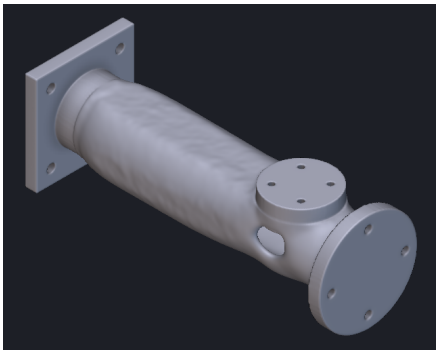
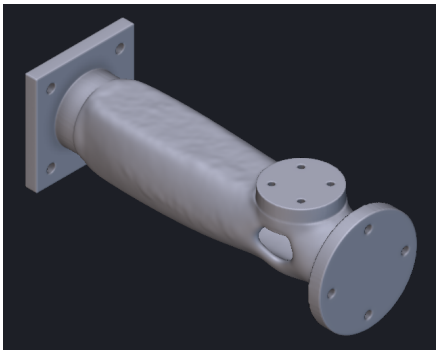
Handling-only	Rotary-only	Dual-mode
		

Figure 8. Original nTop screenshots of the three reconstructed layouts, displayed at the same view and scale. These are geometry views, not measured section-thickness or printability evidence.

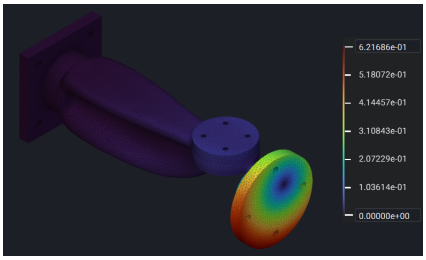
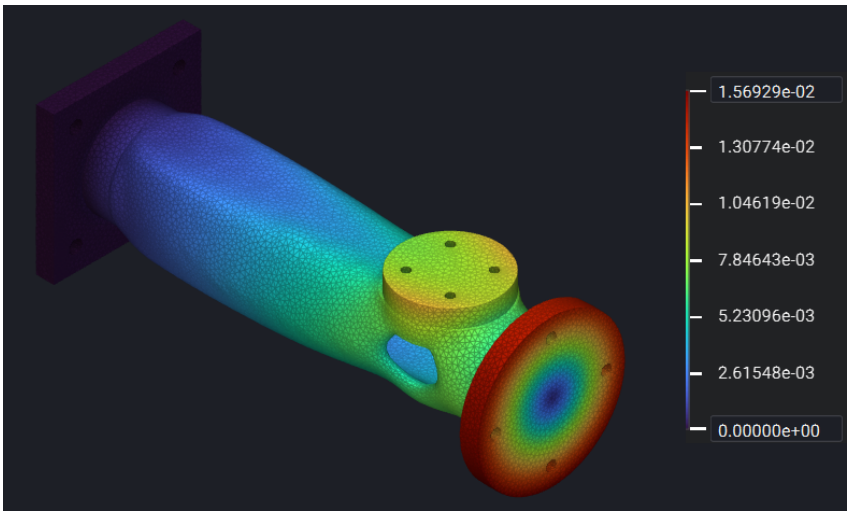
Handling-only, case B	Dual-mode, case B
	

Figure 9. Original nTop total-displacement views under B, in mm. The colour scales differ: handling-only reaches 0.622 mm and dual-mode 0.0157 mm. The handling view is drawn at deformation scale 1 (true scale, so the 0.6 mm is barely visible) and the dual view at scale 0 (undeformed shape); compare the numeric legends, not drawn movement.

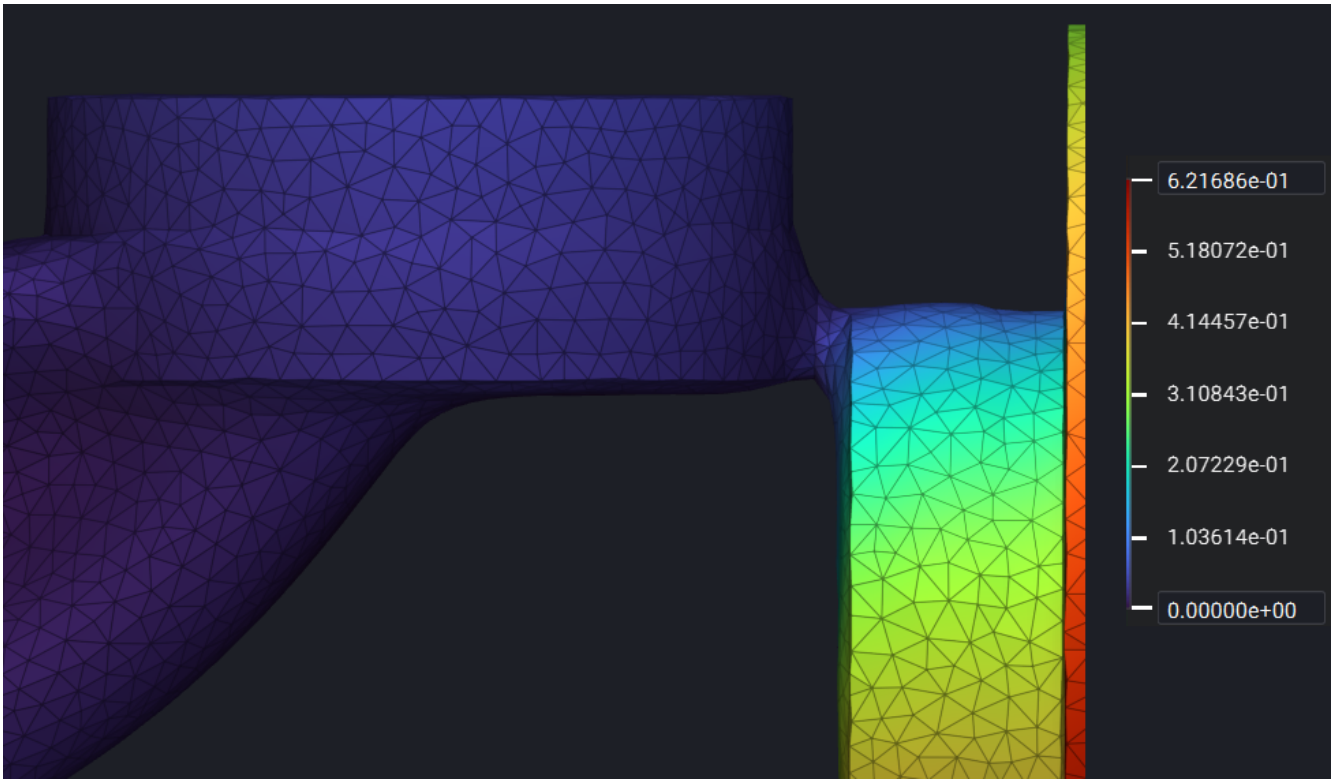


Figure 10. Original nTop close view of handling-only under B. The displacement changes rapidly across the narrow connection toward the hub; its physical thickness and local strength were not measured.

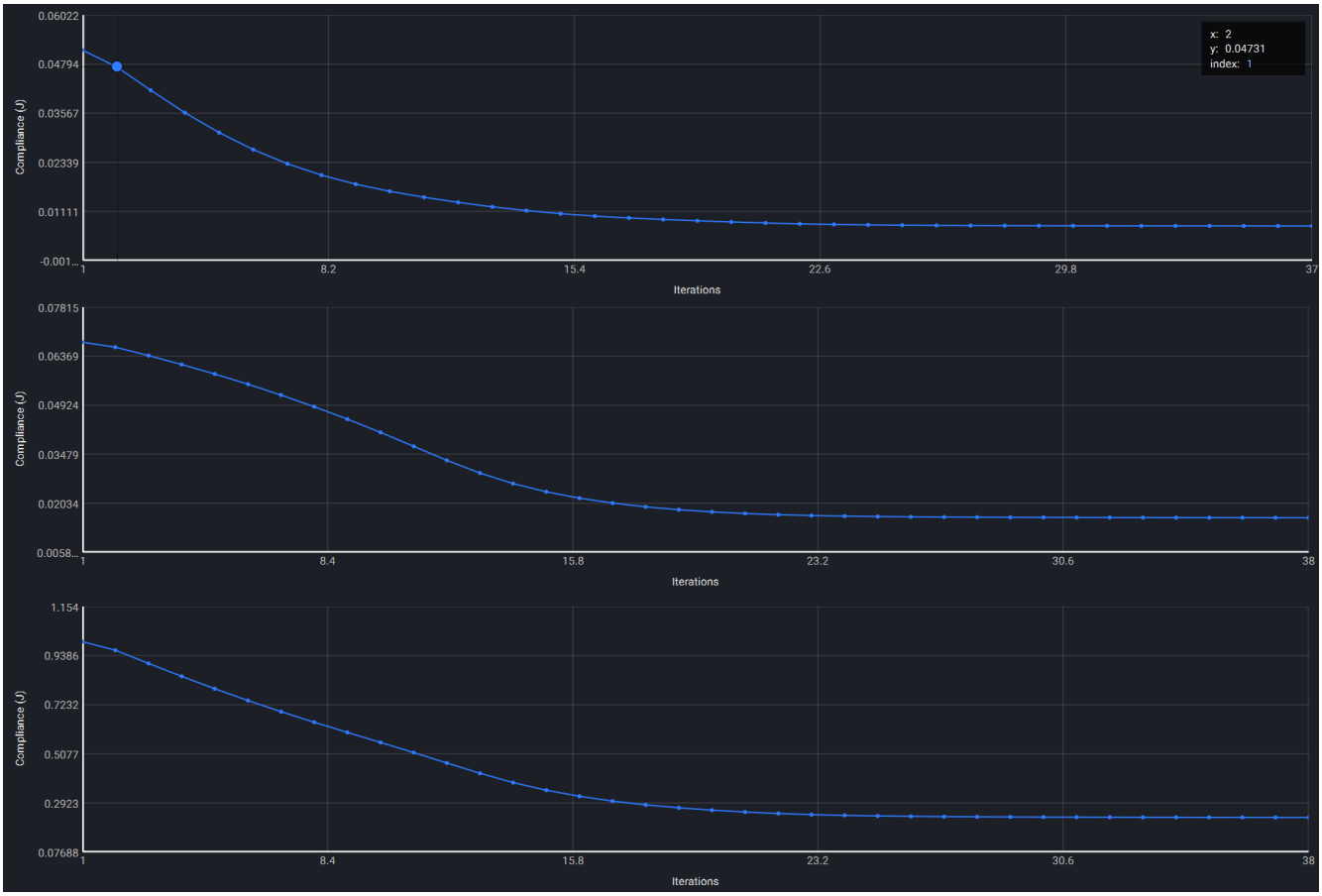


Figure 11. Original nTop objective histories, handling, rotary and dual from top to bottom. The first two plot compliance in J; the weighted dual sum is dimensionless even though its screenshot axis says J. The curves alone do not verify solver termination or constraint feasibility.

8. How the data was evaluated

The current nTop point-map exports are headerless CSVs: six columns for coordinates and vector displacement or reactions, four for coordinates and scalar von Mises stress. The script scripts/analyze_tooling.py interprets coordinates as mm, displacements as m, stress as Pa and reactions as N. It converts displacement to mm for reported movement, stress to MPa, and reaction coordinates to metres for moment calculation. These units are consistent with the saved exports and force equilibrium; an independent native export-dialog record is not included. The script computes:

• the summed root reactions and their moments about the origin, as an equilibrium check;

• the mean vertical displacement of the pad face and the mean axial displacement of the flange face;

• the flange twist, as an unweighted least-squares small-angle rotation about X after removing the selected nodes' mean translations;

• external-work estimates, C_A = 250(−mean pad uz) and C_B = 300(−mean flange ux) + 24θ, using metres and radians in the work terms.

The pad selector uses |z−35|<0.1 mm and radial distance below 26.5 mm about (185,0); the flange selector uses |x−240|<0.1 mm and radius below 40.5 mm about the X axis. These approximate the 0.2 mm load-selector boxes, but equal nodal means and the fitted rotation are not proved identical to the actual force- and connector-work-conjugate displacements. The fit residual is small: the RMS of the in-plane residual vector is 0.00014 to 0.00018 mm, against a rigid tangential displacement of 0.006 to 0.32 mm at 30 mm radius. Widening the plane bands to ±0.3 and ±0.5 mm changes the rotary-versus-dual gap within the reported 0.56–0.65 % range.

For a linear FE system, exact discrete compliance is $C = f^{T}u$, with generalized torque work Mθ included, and strain energy is C/2 under the same load. No exact solver work-conjugate compliance was exported for the eight final cases. The values in Tables 8a and 9 are therefore labeled **estimates**. The Structural Compliance Response used during optimization is a different source and was not applied to all eight final solids for this report. Nodal peak stress is the largest sampled value; the unweighted nodal 99.9th percentile describes distribution but is not a converged material-volume statistic. Masses originate from recorded nTop Mass Properties, not from displacement CSVs. scripts/plot_results.py generates Figures 5–7 from the recorded masses and summary CSV.

The 28 raw CSV files and both scripts are present locally. analyze_tooling.py has hardcoded filenames, accepts an optional raw-data directory, and writes results/tooling_arm_metrics.csv and results/design_summary.csv in the project. plot_results.py writes three PNGs. Rerunning analyze_tooling.py in a separate copy of the project reproduces both summary CSVs byte for byte. The scripts were run with Python 3.10, NumPy 2.2.6 and Matplotlib 3.10.9, the versions pinned in requirements.txt. The intended public file set excludes the raw exports and the 1 GB nTop notebook, so it documents the calculations but is not independently reproducible from included files alone. Appendix C lists the evidence and missing sources.

9. Tools and resources

Table 10. Software, data and references used in the study.

Resource	Used for
Autodesk Fusion	Parametric baseline model (v02) with named parameters; STEP export
nTop 6.0.3	Implicit modelling, preserved regions, FE meshing, static analysis, topology optimization, reconstruction, Mass Properties, point-map export
Python 3.10 with NumPy 2.2.6 and Matplotlib 3.10.9	scripts/analyze_tooling.py (equilibrium, displacement, twist, work estimates, sensitivity); scripts/plot_results.py (Figures 5 to 7)
nTop documentation [3]–[7]	Point Moment and its RBE3 connector, optimization and post-processing workflow, compliance response definition, multi-response weighting, available static-result outputs
Bendsøe and Sigmund [2]	Compliance minimization and the treatment of multiple load cases
Van Cauwenbergh et al. [8]	Yield strength of LPBF AlSi10Mg, as context for the stress peak
Automation.com case study [1]	Industrial precedent for a multi-tool end-of-arm adapter
Own 2D SIMP playground	Building intuition for penalization, filter radius and volume fraction before the 3D study; not used to compute any result here
Study data	28 local nTop point-map exports (110.2 MB decimal) and the local 1.063 GB nTop notebook; excluded from the proposed GitHub package

10. Sources of uncertainty

Table 11. Uncertainties ranked by their effect on the comparison.

Source	Size	Effect on the rotary vs dual ranking
Unequal reconstructed mass	Up to 1.7 % between designs	Can swap the order (k = 1.20)
Linear tetrahedra at 2.5 mm on thin members	Not quantified	Unknown, affects every design
Optimizer stopped above density-change criterion	Not quantified	Unknown
Weight ratio, pilot vs baseline-derived	1.5 %	Small
Interface selection band	Gap varies 0.56 % to 0.65 %	No swap
Partial face coverage on final meshes	77 % to 85 % of nodes	Same for all three designs

The direction and size of the final-body FE discretization error have not been measured. Linear element order, 2.5 mm nominal size, local mesh quality and implicit surface approximation may affect the thin connections differently. No claim about whether the handling torsional penalty is understated or overstated follows from the existing two-mesh baseline check.

11. Limitations and conclusions

The study is linear static with an isotropic benchmark material, a fully fixed rear face and no bolts, contact, preload or joint compliance. The loads are illustrative and do not come from a robot or tool datasheet. No qualified yield, fatigue, buckling or AM process assessment was made. The final meshes were solved at one accepted size only. Neither the native optimizer stop condition nor exact work-conjugate compliance of the final bodies has been independently established.

The supported finding is the pronounced cross-load asymmetry: handling-only has the best estimated A response among lightweight designs but a very weak B response; rotary-only retains useful A response. The dual layout improves A relative to rotary-only but gives up some B response. The rotary-only and dual mean scores are too close, at unequal realized mass, for a reliable overall ranking. The narrow handling connection is a load-path hypothesis and local strength concern, not a proved material failure. The pilot demonstrates

engineering formulation, numerical comparison and critical interpretation; it does not qualify a manufactured arm.

12. Minimum meaningful future work

For engineering feedback on the pilot, no new simulation is required if the close ranking remains explicitly unresolved. Before claiming a strictly controlled study, record the solver stopping state of each run and the smoothing inputs of all three reconstructions. If a decisive rotary-versus-dual ranking becomes a research objective, compare equal-mass final geometries using work-conjugate responses and a targeted final-body mesh sensitivity check. An AM or industrial extension would first require a selected process, actual interface and load data, material allowables and joint modelling; printability and strength cannot be inferred from this study.

Appendix A: Recorded numerical setup and provenance

Item	Recorded value	Verification limit
Baseline CAD	Fusion v02; STEP AP214; STEP SHA-256 6BDB98FD4BF59DC785A32E3C423ABD9A0FD5C067B65F66B4D38B5B0581F4C251	STEP matches supplied copy; Fusion feature history from prior review
nTop	6.0.3; ntop/ToolingArm_Pilot_Completed_v1.ntop	About 1 GB; not in the public repository, available on request
Material	E 70 GPa; ν 0.33; ρ 2700 kg/m³	Recorded benchmark, not an alloy certificate
Optimization FE	2.5 mm linear tetrahedra; 396,426 elements; 71,589 nodes	Node count corroborated by baseline CSV; element count recorded
Common optimization settings	Filter 8 mm; max iterations 100; min objective change 0.0005; min density change 0.01; boundary penalty 0.5; save increment 5	Recorded in config/engineering_inputs.json; solver stopping reason not recorded
Dual weights	wA 9.718173; wB 7.372457	Recorded pilot settings derived from first-iteration histories
Reconstruction	Threshold 0.5; smoothing (dual: 1 mm grid, 3 iterations); zero-radius union with P; intersection with baseline	Smoothing inputs of handling and rotary not recorded side by side
Final meshing	Surface tolerance 0.75 mm; surface and tetra target 2.5 mm; regularity 10; linear FE	Recorded procedure; accepted final meshes solved at one size
Final nodes	Handling 52,635; rotary 55,371; dual 55,073	Corroborated by matching raw export row counts

Appendix B: Eight-case reaction and displacement check

Table B1. Values independently recomputed from the current full-vertex CSVs. Principal root forces and moments are listed with small secondary moments to show that they do not vanish exactly. Reaction moments use coordinates in metres. Max displacement is the largest nodal vector magnitude.

Geometry / case	Nodes	Max displacement (mm)	Principal reaction	Other root moments (N·m)
Baseline A	71,589	0.022772	Fz +249.99961 N; My −46.24955 N·m	Mx −0.00015; Mz −0.00025
Baseline B	71,589	0.008493	Fx +300.00000 N; Mx −24.00000 N·m	My +0.00015; Mz −0.00001
Handling A	52,635	0.033608	Fz +250.00004 N; My −46.21544 N·m	Mx +0.00311; Mz ~0
Handling B	52,635	0.621686	Fx +300.00091 N; Mx −23.99997 N·m	My −0.01465; Mz −0.05555
Rotary A	55,371	0.037387	Fz +250.00001 N; My −46.21609 N·m	Mx −0.01145; Mz ~0
Rotary B	55,371	0.014273	Fx +300.00001 N; Mx −24.00000 N·m	My +0.05965; Mz +0.00600
Dual A	55,073	0.034505	Fz +250.00007 N; My −46.22603 N·m	Mx +0.05962; Mz +0.00018
Dual B	55,073	0.015693	Fx +299.99997 N; Mx −24.00000 N·m	My −0.01644; Mz +0.00582

Appendix C: Evidence, corrections and remaining gaps

The primary available response evidence is results/raw/ (28 point maps); current summaries are results/tooling_arm_metrics.csv and results/design_summary.csv. The calculation definitions are in scripts/analyze_tooling.py; scripts/plot_results.py generates the mass, cross-load and illustrative mass-sensitivity charts. config/engineering_inputs.json records geometry, loads, settings, weights and nTop Mass Properties. Original visual evidence includes the baseline STEP reimport in cad/baseline/review/ and three topology, three objective-history and three deformation/neck screenshots in results/figures/. Figure 2–4 schematics were constructed from the specified geometry and loads; they are not nTop screenshots.

Historical debugging relevant to the accepted result includes a mistaken handling-face selection, a missing torque in an intermediate calculation, a B-only settings mismatch, invalid one-row reaction exports and mesh-reference errors. The current full-vertex reaction files and postprocessing include torque work; the earlier erroneous files and scripts are not present in this workspace. Without versioned notebook run records, the detailed correction sequence and exact final run identities remain narrative evidence. The native Fusion file is kept outside the project folder and is available on request. The nTop notebook and raw exports are local but excluded by .gitignore; final optimized STEP bodies, a Git history, an AM build record and physical test data are absent.

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